Introduction to Network

What is Computer Network?

A group of computer **which are connected to each – other** for the purpose of **sharing their resources** is called computer network.

First Computer Network – ARPANET

The U.S. Advanced Research Projects Agency Network (ARPANET) was **the first public packet-switched computer network**. It was first used in 1969

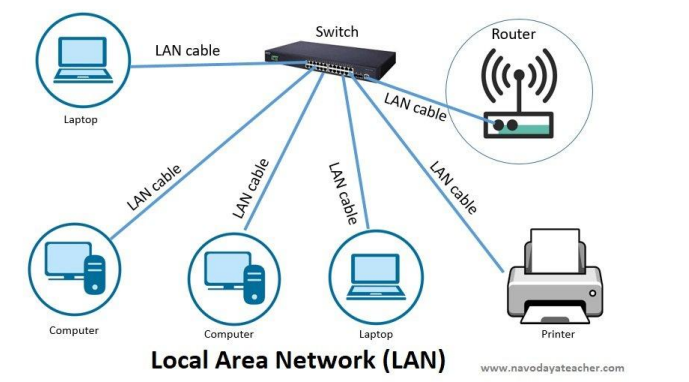
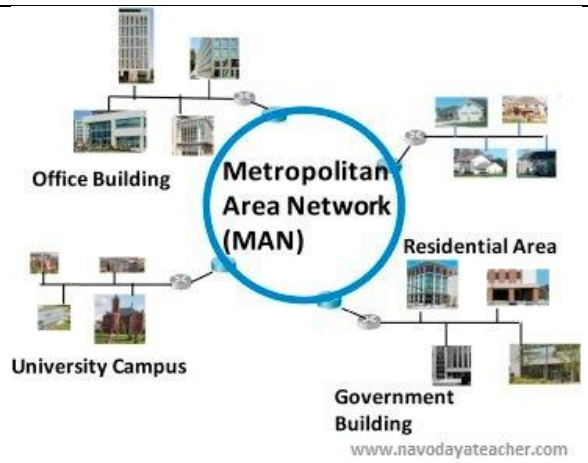
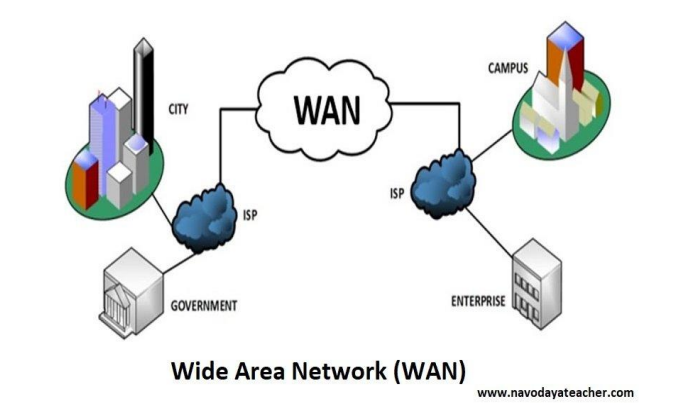
Characteristics of Computer Network

1. Resource Sharing
2. Communication Speech
3. Backup
4. Scalability
5. Reliability
6. Software & Hardware Sharing
7. Security

Network Devices

HUB, Switch, Bridge, Gateway, Modem. Router, Repeater etc.

Network Types

	
1. LAN (Local Area Network) • Range : <150m • Use – Office (Building)	2. MAN (Metropolitan Area Network) • Range : <50km • Use – Within City
	
3. PAN (Personal Area Network) • Range : <10m • Use – Home (For Personal Use)	4. WAN (Wide Area Network) • Range : Not Fixed • Use – Use for Countries or All around world.

Advantages :

1. Open to Everyone
2. File sharing
3. Security
4. Easy to add new devices
5. Backup Storage.

Disadvantage :

1. Network device required
2. Virus attack
3. Required Hander
4. High Speed Internet
5. Server

Difference Between LAN, MAN & WAN in Computer Networks

LAN	MAN	WAN
LAN stands for Local Area Network	MAN stands for Metropolitan Area Network	WAN stands for Wide Area Network
It is used for building like offices.	It is used for city like Ahmedabad.	It is used for countries.
Transmission speed of data is high.	Transmission speech of data is average.	Transmission speed of data is low.
LAN network range is 0 to 150m	MAN network range is 5 to 50km	Not fixed
LAN Network ownership is private	MAN network ownership is Private or Public	WAN network ownership is also Private or Public
Easy to Maintain	Difficult to maintain than LAN	Also difficult to maintain than MAN as well as LAN
LAN network error rate & setup cost is low	MAN network error rate & setup cost is average	WAN network error rate & setup cost is very high.

What is Network Cabling?

Network Cables: To connect two or more computers or networking devices in a network, network cables are used.

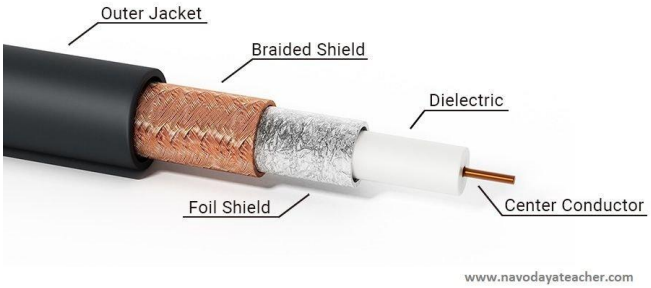
Types of Cables used in Networks

1. Coaxial Cable
2. Twisted Pair Cables
3. Fiber Optic Cable

1. Coaxial Cable

This cable contains a conductor, insulator, braiding & sheath.

Coaxial Cable Example



Coaxial Cable Conductor types (1. Thick Coaxial – 10Base5 – 2-5mm thick) (2. Thin Coaxial (10Base2 – 6mm thick)

10BaseT : 10 = 10 MBPS, Base = Half Duplex, T = Transmission Distance

Categories of Coaxial Cables

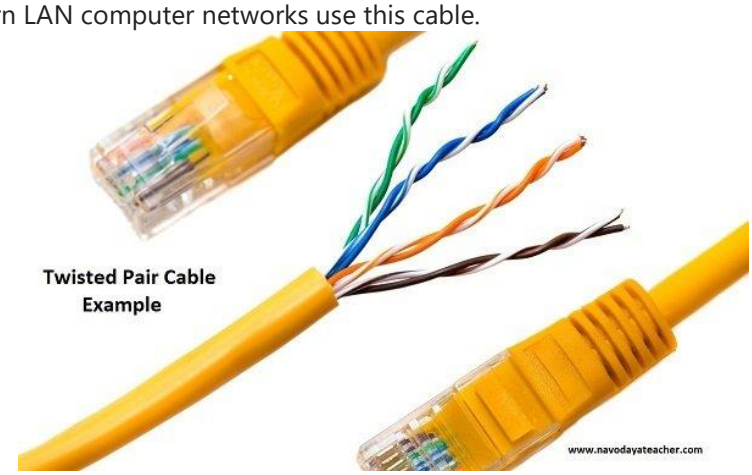
Types	Impedance	Used
RG-59	75	Short-distance cable TV
RG – 58	50	Thin Ethernet Cable
RG – 11	50	Thick Ethernet Cable

Applications of Coaxial Cables

1. Analog telephone networks
2. Digital Telephone networks
3. Cable TV networks
4. Traditional Ethernet LANs

2. Twisted-pair cables

It was primarily developed for computer networks. This cable is also known as Ethernet Cable. almost all modern LAN computer networks use this cable.



TIA / EIA specifies standards for the twisted-pair cable. First standards were released in 1991. Known as TIA/EIA 568.

These standards have been continually revised to cover the latest technologies and developments of the transmission media.

Twisted-pair cables

1. UTP – Unshielded twisted-pair (UTP Cables)
2. STP – Shielded twisted-pair (STP Cables)

Connector Used –

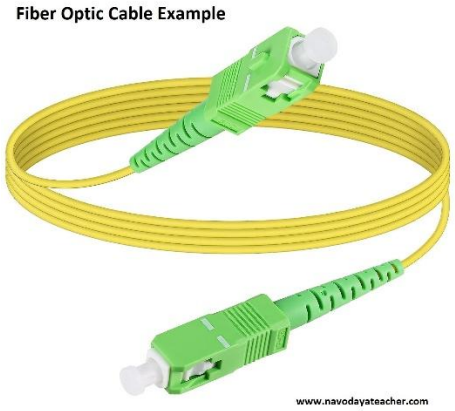
Perticulars	RJ-45	RJ-11
Application	Ethernet Cables	Connecting telephone wires
Number of wires	8	4
Size	Big	Small
Bandwidth	10 GBPS over Ethernet	24 MBPS

3. Fiber Optic Cable / Optical Fiber Cable

This cable consists of core, cladding, buffer, and jacket.

The core is wrapped in the cladding; Cladding is wrapped in the buffer, and the buffer is wrapped in the jacket.

Structure of Fiber Optics



Version 1.0.0

This cable can transmit data over a long distance at the highest speed. It can transmit data up to 40 Kilometers at the speed of 100 GBPS.

Advantages of Fiber Optic Cables

- Fiber optics support a higher capacity (Fiber Cables rated at 10Gbps, 40Gbps, and 100 Gbps are standard)
- Transmit signals over much longer distances

Two common types of fiber cables

1. Single mode Fiber (SMF)
2. Multi mode giber (MMF)

Multimode cable has a larger diameter. Single mode can provide more distance, but it is more expensive.

Introduction to Internet

The Internet –

The Internet is the global system of interconnected computer networks that uses the Internet protocol suite (TCP/IP) to communicate between networks and devices.

The Internet is used for many things, such as electronic mail, online chat, file transfer, and the interlinked web pages and other documents of the World Wide Web.

What is Internet ?

Internet is a network of networks that is used to interlinked many different types of computers all over the world.

The Internet is a vast network that connects computers all over the world. Through the Internet people can share information and communicate from anywhere with an internet connection.

Application of Internet

1. Education
2. E-mail
3. Business
4. E-commerce
5. Media and Entertainment
6. Social Networking
7. Forum
8. Health and Fitness

Disadvantage of Internet

1. Virus Threat
2. Spamming
3. Cyber Crime
4. Cyber Terrorism
5. Time Wasting

World Wide Web

The World Wide web ('www' or 'web' for short) is a collection of webpages found on this network of computers. your web browser uses the internet to access the web.

Web Browsers

A web browser, or simply 'browser'. is an application used to access and view websites. Common web browsers include Microsoft Internet Explorer (Ended on June 15, 2022) Now Microsoft Edge, Google Chrome, Mozilla Firefox, Opera and Apple Safari.

Website

A website is a collection of web pages and related content that is identified by a common domain name and published on at least one web server.

Web Page

A web page (or webpage) is a specific collection of information provided by a website and displayed to a user in a web browser.

There are two types of Web pages:

1. Static web pages
2. Dynamic web pages

Protocol

A protocol is a standard set of rules that allow electronic devices to communicate with each other. These rules include what type of data may be transmitted. what commands are used to send and receive data, and how data transfers are confirmed.

Example:

1. HTTP
2. FTP
3. TCP
4. IP
5. ARP

Home Page

A Home Page (also written as homepage) is the main web page of a website The term can also refer to one or more pages always shown in a web browser when the application starts up. In this case, it is also known as the start page.

URL (Uniform Resource Locator)

A URL is the location of a webpage or file on the internet. e.g. https://www.studmentor.in

The URL has two parts

1. Protocol Identifier: https://
2. Resource name: www.studmentor.com

What is Search Engine?

A Search Engine is a software system designed to search out the information from world wide web according to the user's query.

Example of Search Engines:

1. Google
2. Microsoft Bing
3. Yahoo!
4. DuckDuckGo
5. Baidu
6. Ask.com

World Wide Web

- The WWW (World Wide Web) or simply web is a means of accessing information over the internet.
- It contains millions of documents or 'bunch of pages' called web pages.
- Each page contains text, graphics, video, audio and is linked to other web pages via hyperlink.
- The web uses the HTTP (Hyper Text Transfer Protocol) for the transmission of these documents over the Internet.

Saving Information from Internet

Downloading

Usually, when you download a file you will start the download by clicking a link to that file.

1. Most files: Click the download link. You can also right-click on the file and choose Save as.
2. Images: Right-click on the image and choose Save Image As.
3. Videos: Point to the video. Click Download

Working with Excel

What is MS Excel?

- Microsoft Excel is a software program developed by Microsoft
- It allows users to organize, format and calculate data with formulas using spreadsheet system.
- Microsoft Excel is a helpful and powerful program for data analysis and documentation.
- By Default, MS -Excel contains three worksheets.
- It is a part of MS Office.
- It is also known as electronic spreadsheet.

What is Worksheet?

- A worksheet is a collection of rows and columns.
- Worksheet is a component of workbook
- When a row and a column meet, they form a cell.
- Cells are used to record data.
- Each cell is uniquely identified using a cell address.
- Columns are usually labelled with letters while rows are usually numbers.
- A Spreadsheet/Worksheet contains 1,048,576 rows and 16384 column.
- It is also known as Spreadsheet in Excel.

What is Workbook?

- A workbook is a collection of worksheets.
- By default, a workbook has three worksheets in Excel (Variant in Versions)
- You can delete or add more sheets as per your requirements.
- You can rename the sheet names to more meaningful. i.e. Daily Expanses, Monthly Budget etc.

Steps to Start Excel

How to open MS Excel?

To open MS Excel on your computer, follow the steps given below:

- Click on Start
- Then All Programs
- Next step is to click on MS Office

- Then finally, choose the MS-Excel option

Components of Excel Screen

1. Workbook Tab
2. Status Bar
3. Worksheet
4. Scroll Bars
5. Formula Bar
6. Name Box
7. Ribbon
8. Display Options
9. Quick Access Toolbar
10. Title Bar
11. Active Cell
12. Zoom Slider
13. Close Button

Table Creating in MS Excel

(a) Cell reference in MS Excel

A cell reference in Microsoft Excel is a combination of a column letter and a row number that identifies a cell or range of cells on a worksheet. Cell references can be used in formulas to help Excel find the data or values to calculate.

(b) Functions in MS Excel

Functions and Formulas are the backbone of your Excel skills. If you are comfortable in using just a few powerful functions and formulas you are already ahead of many people around you. That is a good reason to gain some knowledge about some basic functions and formulas you should know for your everyday tasks.

The Difference between an Excel Function and Formula

Function is a piece of code designed to calculate certain values. A formula on the other hand can contain values, references to cells, and functions. So that means, if you are calculating something in Excel you use formulas and use functions within your formulas.

Now, let's have a look at the ten most fundamental functions to create powerful formulas in Excel.

1. SUM Function – **Syntax:** SUM (number1, [number2], [number3], ...)
2. AVERAGE Function – **Syntax:** AVERAGE (number1, [number2], [number3], ...)
3. IF Function – **Syntax:** IF (logical_test, [value_if_true], [value_if_false])
4. MIN & MAX Function – **Syntax:** MIN (number1, [number2], ...) | MAX (number1, [number2], ...)
5. TRIM Function – **Syntax:** TRIM (text)
6. CONCATENATE Function – **Syntax:** CONCATENATE (text1, [text2], ...)
7. COUNT Function – **Syntax:** COUNT (value1, [value2], ...)
8. COUNTA Function – **Syntax:** COUNTA (value1, [value2], ...)
9. COUNTIF Function – **Syntax:** COUNTIF (range, criteria)
10. SUMIF Function – **Syntax:** SUMIF (range, criteria, [sum_range])

Working with Powerpoint

1. What is Power Point ?
2. Insert a new slide
3. Insert a image
4. Word art
5. Audio and Video
6. Transitions in slide
7. Custom animations

- **What is PPT?**

  - PPT stands for **PowerPoint Presentation**, created using Microsoft PowerPoint.
  - It is a tool for creating **visual presentations** with text, images, charts, audio, and video.

- **Features of PPT:**

  - **Slides**: Basic units of a presentation.
  - **Templates & Themes**: Pre-designed layouts for quick presentations.
  - **Animations**: Effects applied to text, images, or objects.
  - **Transitions**: Special effects between slides.
  - **Multimedia Support**: Add audio, video, and images.
  - **Charts & SmartArt**: Visual representation of data.

- **Uses of PPT:**

  - Teaching and learning.
  - Business meetings & reports.
  - Project presentations.
  - Training & workshops.
  - Event and seminar displays.

- **Advantages:**

  - Easy to use and visually attractive.
  - Helps in **clear communication**.
  - Saves time with templates.
  - Engages audience with multimedia.

- **Tips for Good PPT:**

  - Keep slides **simple and clear**.
  - Use **bullet points** instead of long text.
  - Add relevant images, graphs, and charts.
  - Limit animations to avoid distraction.
  - Use **readable fonts** and proper color contrast.